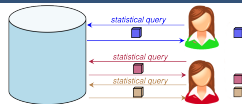


Macrodata and Microdata Protection

Security, Privacy, and Data Protection Laboratory
Dipartimento di Informatica
Università degli Studi di Milano

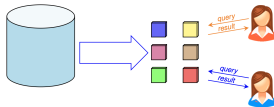
Outline

- Statistical DBMS



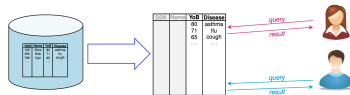
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Statistical data dissemination

- Often **statistical data** (or data for statistical purpose) are released
- Such released data can be used to infer information that was not intended for disclosure
- Disclosure can:
 - occur based on the released data alone
 - result from combination of the released data with publicly available information
 - be possible only through combination of the released data with detailed external (public) data sources
- The **disclosure risk** from the released data should be very low

Statistical DBMS vs statistical data

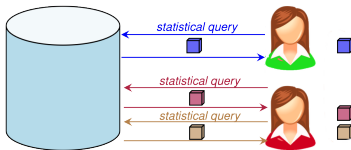
Release of data for statistical purpose

- statistical DBMS
 - the DBMS responds only to statistical queries
 - need run time checking to control information (indirectly) released
- statistical data
 - publish statistics (macrodata release)
 - control on indirect release performed before publication

Statistical DBMS

Statistical DBMS

- A **statistical DBMS** is a DBMS that provides access to **statistics** about **groups** of individuals
 - should **not** reveal information about any **particular individual**
- Confidential information about an individual can be deduced
 - combining the results of **different statistics**
 - combining the results of statistics with **external knowledge** (possibly about the database content)



Statistical DBMS – Example (1)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query: sum of the incomes of females with major in EE

Statistical DBMS – Example (1)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query: sum of the incomes of females with major in EE

Result: it reveals the income of Baker (only female with EE)

Statistical DBMS – Example (1)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query: sum of the incomes of females with major in EE

Result: it reveals the income of Baker (only female with EE)

⇒ The query is sensitive

Statistical DBMS – Example (1)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query: sum of the incomes of females with major in EE

Result: it reveals the income of Baker (only female with EE)

⇒ The query is sensitive

⇒ Block statistics computed over a too small number of respondents

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 1: sum of the incomes of individuals with major in EE

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 1: sum of the incomes of individuals with major in EE

Result: it does not reveal the income of any individual (240k)

⇒ The query is not sensitive

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 2: sum of the incomes of males with major in EE

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 2: sum of the incomes of males with major in EE

Result: it does not reveal the income of any individual (190k)

⇒ The query is not sensitive

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 1: sum of the incomes of individuals with major in EE (240k) —

Query 2: sum of the incomes of males with major in EE (190k)

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 1: sum of the incomes of individuals with major in EE (240k) –

Query 2: sum of the incomes of males with major in EE (190k)

= sum of the incomes of females with major in EE (50k)
income of Baker

Statistical DBMS – Example (2)

Name	Sex	Major	Class	Income
Allen	Female	CS	1980	68k
Baker	Female	EE	1980	50k
Cook	Male	EE	1978	70k
Davis	Female	CS	1978	80k
Evans	Male	EE	1981	60k
Frank	Male	CS	1978	76k
Good	Male	CS	1981	64k
Hall	Male	EE	1978	60k
Iles	Male	CS	1979	70k

Query 1: sum of the incomes of individuals with major in EE (240k) –

Query 2: sum of the incomes of males with major in EE (190k)

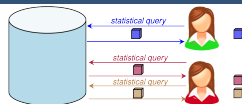
= sum of the incomes of females with major in EE (50k)
income of Baker

⇒ The combination of queries is sensitive

Macrodata protection

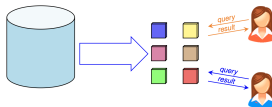
Outline

- Statistical DBMS



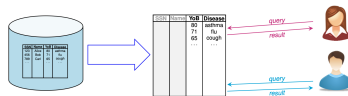
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

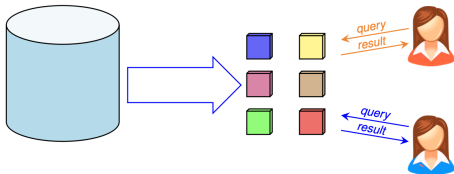
- Masking techniques
- Synthetic techniques



Macrodata

Macrodata tables can be classified into the following two groups (types of tables)

- **Count/Frequency.** Each cell contains the number (count) or the percentage (frequency) of respondents that have the same value over all attributes in the table
- **Magnitude data.** Each cell contains an aggregate value of a *quantity of interest* over all attributes in the table



Count table – Example

Two-dimensional table showing the number of employees by department and annual income (in K Euro)

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁	2	4	18	20	7	1	52
Dept ₂	-	-	7	9	-	-	16
Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Magnitude table – Example

Average number of days spent in the hospital by respondents with a given disease

	Hypertension	Obesity	Chest Pain	Short Breath	Tot
M	2	8.5	23.5	3	37.0
F	3	30.5	0	5	38.5
Tot	5	39.0	23.5	8	75.5

Microdata table – Example

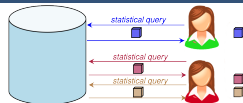
Records about employees of company Alfa

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Pete	Alfa	high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmi	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	Mike	Alfa	low	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Macrodata Disclosure Protection Techniques: Tables of Counts or Frequencies

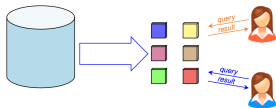
Outline

- Statistical DBMS



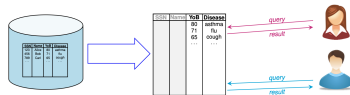
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Tables of counts or frequencies

- Data collected from most surveys are published in tables of count or frequencies
- Protection operates in three steps
 1. Sampling
 2. Identification of sensitive cells
 - special rules
 - threshold rules
 3. Protection of sensitive cells
 - table restructuring
 - suppression
 - rounding
 - confidentiality edit

Sampling

- Conduct (and publish) a **sample survey** rather than a census
- Estimates are made by multiplying individual responses by a **sampling weight** before aggregating them
- If weights are not published, weighting helps to make an individual respondent's data less identifiable from published totals
- Estimates must achieve a specified **accuracy**
 - data that do not meet the accuracy requirements are not published (not considered meaningful)

Tables of counts or frequencies

- Data collected from most surveys are published in tables of count or frequencies
- Protection operates in three steps
 1. Sampling
 2. Identification of sensitive cells
 - special rules
 - threshold rules
 3. Protection of sensitive cells
 - table restructuring
 - suppression
 - rounding
 - confidentiality edit

Special rules

- When macrodata tables are defined on the whole population, disclosure limitation procedures must be applied
- **Special rules** define restrictions on the level of detail that can be provided in a table
- Special rules differ depending on the **agency** and the **kind of table**

Special rules – Example (1)

Social Security Administration (SSA) rules prohibit publishing tables where the value of a cell:

- is equal to a marginal total or
- would allow users to determine
 - an individual's age within a five-year interval
 - earnings within a \$1,000 interval
 - benefits within a \$50 interval
- to satisfy special rules
 - table restructuring or category combination

Special rules – Example (2)

Number of employees by department and annual income (in K Euro)

Special rule: Income within a 5K Euro interval

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁	2	4	18	20	7	1	52
Dept ₂	-	-	7	9	-	-	16
Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Special rules – Example (2)

Number of employees by department and annual income (in K Euro)

Special rule: Income within a 5K Euro interval

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
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Dept ₂	-	-	7	9	-	-	16
Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Cells that cannot be released

-  its value is equal to total

Special rules – Example (2)

Number of employees by department and annual income (in K Euro)

Special rule: Income within a 5K Euro interval

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
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Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Cells that cannot be released

-  its value is equal to total
- , ,  allow recipients to determine income within a 5K interval

Special rules – Example (2)

Number of employees by department and annual income (in K Euro)

Special rule: Income within a 5K Euro interval

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁	2	4	18	20	7	1	52
Dept ₂	-	-	7	9	-	-	16
Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Cells that cannot be released

-  its value is equal to total
- , ,  allow recipients to determine income within a 5K interval
 -  between 23K and 25K for Dept₄
 - ,  between 23K and 27K for Dept₂

Special rules – Example (U.S. HIPAA)

Health Insurance Portability and Accountability Act

“Safe Harbor” rules, include:

- identifying information must be removed
- locations have to be generalized to units that contain at least 20,000 residents
- dates of birth must be rounded up to the year of birth only (or to larger value if the person is older than 90)

Threshold rules

- A cell is sensitive if the number of respondents is less than some specified number (e.g., some agencies consider 5, others 3)
- A sensitive cell cannot be released
- Different techniques can be applied to protect sensitive cells:
 - table restructuring and category combination
 - cell suppression
 - random rounding
 - controlled rounding
 - confidentiality edit

Table with disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppress one additional cell for each row/column with a sensitive cell suppressed

Table with disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
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Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

A cell with fewer than **5** respondents is defined as sensitive
Suppress one additional cell for each row/column with a sensitive cell suppressed

Table with disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
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A cell with fewer than **5** respondents is defined as sensitive
Suppress one additional cell for each row/column with a sensitive cell suppressed

Tables of counts or frequencies

- Data collected from most surveys are published in tables of count or frequencies
- Protection operates in three steps
 1. Sampling
 2. Identification of sensitive cells
 - special rules
 - threshold rules
 3. Protection of sensitive cells
 - table restructuring
 - suppression
 - rounding
 - confidentiality edit

Table restructuring – Example

- Number of employees by department and annual income (in K Euro)
Special rule: Income within a 5K Euro interval
- To protect confidentiality, the table can be restructured and rows or columns combined (“rolling-up categories”)

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁	2	4	18	20	7	1	52
Dept ₂	-	-	7	9	-	-	16
Dept ₃	-	6	30	15	4	-	55
Dept ₄	-	-	2	-	-	-	2

Table restructuring – Example

- Number of employees by department and annual income (in K Euro)
Special rule: Income within a 5K Euro interval
- To protect confidentiality, the table can be **restructured** and rows or columns **combined** (“rolling-up categories”)

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁ Dept ₂	2	4	25	29	7	1	68
Dept ₃ Dept ₄	-	6	32	15	4	-	57

- Combining Dept₁ with Dept₂ and Dept₃ with Dept₄ does offer the required protection

Table restructuring – Example

- Number of employees by department and annual income (in K Euro)
Special rule: Income within a 5K Euro interval
- To protect confidentiality, the table can be **restructured** and rows or columns **combined** (“rolling-up categories”)

Dept	Income						Total
	[0-21)	[21-23)	[23-25)	[25-27)	[27-29)	29+	
Dept ₁	2	4	18	20	7	1	52
Dept ₂ Dept ₄	-	-	9	9		-	16
Dept ₃	-	6	30	15	4	-	55

- Combining Dept₂ with Dept₄ would still reveal that the income is within a 5K interval [23K, 27K)

Cell suppression

- One of the most used ways of protecting sensitive cells is suppression
- Suppressing sensitive cells (primary suppression) is not sufficient
- At least one additional cell must be suppressed (complementary suppression) for each row or column with a suppressed sensitive cell (primary suppression)
 - the value in the sensitive cell can be calculated from the marginal total
- Even with complementary suppression it is difficult to guarantee adequate protection

Complementary suppressions

- The selection of cells for complementary suppression is complicated
- **Linear programming** techniques are used to automatically select cells for complementary suppression
- **Audit techniques** can be applied to evaluate the proposed suppression pattern to see if it provides the required protection

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

A cell with fewer than **5** respondents is defined as sensitive

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D₁	D₂	D₃	20
Beta	20	10	10	15	55
Gamma	D₄	10	10	D₅	25
Delta	12	14	7	D₆	35
Total	50	35	30	20	135

Suppress sensitive cells

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	10	10	15	55
Gamma	D_4	10	10	D_5	25
Delta	12	14	7	D_6	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$35 = D_1 + 10 + 10 + 14$$

$$\Rightarrow D_1 = 1$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	D₂	D₃	20
Beta	20	10	10	15	55
Gamma	D₄	10	10	D₅	25
Delta	12	14	7	D₆	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$35 = \mathbf{D_1} + 10 + 10 + 14$$

$$\Rightarrow \mathbf{D_1 = 1}$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	D_2	D_3	20
Beta	20	10	10	15	55
Gamma	D_4	10	10	D_5	25
Delta	12	14	7	D_6	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$30 = D_2 + 10 + 10 + 7$$

$$\Rightarrow D_2 = 3$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D₃	20
Beta	20	10	10	15	55
Gamma	D₄	10	10	D₅	25
Delta	12	14	7	D₆	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$30 = \mathbf{D_2} + 10 + 10 + 7$$

$$\Rightarrow \mathbf{D_2 = 3}$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D ₃	20
Beta	20	10	10	15	55
Gamma	D ₄	10	10	D ₅	25
Delta	12	14	7	D ₆	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$50 = 15 + 20 + D_4 + 12$$

$$\Rightarrow D_4 = 3$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D ₃	20
Beta	20	10	10	15	55
Gamma	3	10	10	D ₅	25
Delta	12	14	7	D ₆	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$50 = 15 + 20 + D_4 + 12$$

$$\Rightarrow D_4 = 3$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D_3	20
Beta	20	10	10	15	55
Gamma	3	10	10	D_5	25
Delta	12	14	7	D_6	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$35 = 12 + 14 + 7 + D_6$$

$$\Rightarrow D_6 = 2$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D_3	20
Beta	20	10	10	15	55
Gamma	3	10	10	D_5	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$35 = 12 + 14 + 7 + D_6$$

$$\Rightarrow D_6 = 2$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	D_3	20
Beta	20	10	10	15	55
Gamma	3	10	10	D_5	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$20 = 15 + 1 + 3 + D_3$$

$$\Rightarrow D_3 = 1$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	D ₅	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$20 = 15 + 1 + 3 + D_3$$

$$\Rightarrow D_3 = 1$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	D_5	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$25 = 3 + 10 + 10 + D_5$$

$$\Rightarrow D_5 = 2$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Suppressing sensitive cells is not sufficient

$$25 = 3 + 10 + 10 + D_5$$

$$\Rightarrow D_5 = 2$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D₁	D₂	D₃	20
Beta	20	D₄	D₅	15	55
Gamma	D₆	10	10	D₇	25
Delta	D₈	14	7	D₉	35
Total	50	35	30	20	135

Suppress one additional cell for each row/column with a sensitive cell suppressed

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D₁	D₂	D₃	20
Beta	20	D₄	D₅	15	55
Gamma	D₆	10	10	D₇	25
Delta	D₈	14	7	D₉	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D₁	D₂	D₃	20
Beta	20	D₄	D₅	15	55
Gamma	D₆	10	10	D₇	25
Delta	D₈	14	7	D₉	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells, but:

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells, but:

$$(15 + D_1 + D_2 + D_3) +$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells, but:
 $(15 + D_1 + D_2 + D_3) + (20 + D_4 + D_5 + 15) -$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells but:

$$(15 + D_1 + D_2 + D_3) + (20 + D_4 + D_5 + 15) - (D_1 + D_4 + 10 + 14) -$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells but:

$$(15 + D_1 + D_2 + D_3) + (20 + D_4 + D_5 + 15) - (D_1 + D_4 + 10 + 14) - (D_2 + D_5 + 10 + 7)$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	D_3	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells but:

$$(15 + D_1 + D_2 + D_3) + (20 + D_4 + D_5 + 15) - (D_1 + D_4 + 10 + 14) - (D_2 + D_5 + 10 + 7) = 20 + 55 - 35 - 30$$

$$\Rightarrow D_3 = 1$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D_1	D_2	1	20
Beta	20	D_4	D_5	15	55
Gamma	D_6	10	10	D_7	25
Delta	D_8	14	7	D_9	35
Total	50	35	30	20	135

The table appears to offer protection to the sensitive cells but:

$$(15 + D_1 + D_2 + D_3) + (20 + D_4 + D_5 + 15) - (D_1 + D_4 + 10 + 14) - (D_2 + D_5 + 10 + 7) = 20 + 55 - 35 - 30$$

$$\Rightarrow D_3 = 1$$

Cell suppression: Table without disclosures – Example

Table containing information about employees by company and education level

Company	Education level				Total
	Low	Medium	High	Very High	
Alpha	15	D ₁	D ₂	D ₃	20
Beta	20	10	10	15	55
Gamma	D ₄	D ₅	10	D ₆	25
Delta	D ₇	14	D ₈	D ₉	35
Total	50	35	30	20	135

The table provides adequate protection for the sensitive cells but out of a total of 16 cells, only 7 cells are published, while 9 are suppressed

Rounding

To reduce data loss due to suppression, use **rounding of values** to a multiple of the sensitivity threshold

- **random**: random decision on whether cell values will be rounded up or down
 - the sum of the values in a row/column may be different from the published marginal totals (recipients may lose confidence in the data)
- **controlled**: ensure that the sum of published entries is equal to published marginal totals

Note: all cell values must be a **multiple** of the threshold value

Random rounding – Example

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Random rounding – Example

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Company	Education level (random rounding)				Total
	Low	Medium	High	Very High	
Alfa	15	*0	*0	*0	20
Beta	20	10	10	15	55
Gamma	*5	10	10	*0	25
Delta	*15	*15	*10	*0	35
Total	50	35	30	20	135

Controlled rounding – Example

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Controlled rounding – Example

Company	Education level				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Company	Education level (controlled rounding)				Total
	Low	Medium	High	Very High	
Alfa	15	*0	*5	*0	20
Beta	20	10	10	15	55
Gamma	*5	10	10	*0	25
Delta	*10	*15	*5	*5	35
Total	50	35	30	20	135

Controlled rounding

- Linear programming methods are used to identify a controlled rounding for a table
- Disadvantages:
 - it requires the use of specialized computer programs
 - controlled rounding solutions **may not always exist** for complex tables

Confidentiality edit (1)

- Developed by the U.S. Census Bureau to provide protection of tables prepared from the 1990 Census
- Two different approaches:
 - to protect the regular decennial Census data (100% of the population)
 - to protect the long-form of the Census which refers to a sample of the population
- Both approaches apply statistical disclosure limitation techniques to the microdata on which statistics are calculated:
 - statistics are protected by changing input data

Confidentiality edit (2)

- For the 100 percent microdata file, confidentiality edit applies switching
 1. Take a **sample** of records from the microdata file
 2. Find a **match** for these records in some other geographic region, matching on a specified set of important attributes
 3. **Swap** all attributes on the matched records
- For small blocks, the sampling fraction is increased to provide additional protection
- The microdata file can be used directly to prepare macrodata tables

Confidentiality edit – Example (1)

Records for the 20 employees of company Alfa

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Pete	Alfa	high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmi	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	Mike	Alfa	low	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Confidentiality edit – Example (2)

1. Take a sample of records from the microdata file (say a 10% sample, 2 tuples for company Alfa). Assume that records number 4 and 17 were selected as part of our 10% sample
2. Since we need tables by company and education level, we find a match in some other company on the other variables (race and salary, company totals for these variables remain unchanged)
 - A match for record 4 (Pete) is found in company Beta, the match is with Alonso, who has very high education
 - Record 17 (Mike) is matched with George in company Delta, who has medium education

Confidentiality edit – Example (3)

3. We also assume that part of the randomly selected 10% sample from other companies match records in company Alfa
 - One record from company Delta (June with high education) matches with Virginia (record 12)
 - One record from company Gamma (Heather with low education) matched with Nancy (record 20)
4. After all matches are made, swap attributes on matched records
5. Use the swapped data file directly to produce tables

Confidentiality edit – Example (4)

Records for the 20 employees of company Alfa

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Pete	Alfa	high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmi	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	Mike	Alfa	low	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Confidentiality edit – Example (4)

Take a sample of records from the microdata file (say a 10% sample)

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Pete	Alfa	high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irm	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	Mike	Alfa	low	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Confidentiality edit – Example (4)

Since we need tables by company and education level, we find a match in some other company on the other variables

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Alonso	Alfa	very high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmie	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	George	Alfa	medium	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Confidentiality edit – Example (4)

Part of the randomly selected 10% sample from other companies match records in company Alfa

N	Employee	Company	Education	Salary	Race
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Alonso	Alfa	very high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmie	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	June	Alfa	high	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	George	Alfa	medium	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Heather	Alfa	low	37	white

Confidentiality edit – Example (5)

Company	Education level (original)				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Confidentiality edit – Example (5)

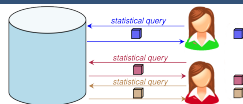
Company	Education level (original)				Total
	Low	Medium	High	Very High	
Alfa	15	1	3	1	20
Beta	20	10	10	15	55
Gamma	3	10	10	2	25
Delta	12	14	7	2	35
Total	50	35	30	20	135

Company	Education level (with confidentiality edit)				Total
	Low	Medium	High	Very High	
Alfa	13	2	3	2	20
Beta	20	10	10	15	55
Gamma	4	9	10	2	25
Delta	13	14	7	1	35
Total	50	35	30	20	135

Macrodata Disclosure Protection Techniques: Tables of Magnitude Data

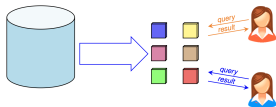
Outline

- Statistical DBMS



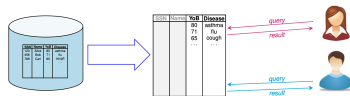
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Protection of tables of magnitude data – 1

- Magnitude data are generally **nonnegative quantities** reported in surveys or censuses
- The distribution of these values is likely to be **skewed**
- Disclosure limitation techniques focus on preventing precise estimation of the values for **outliers**
- Sampling is less likely to provide protection
- The units that are most visible because of their size do not receive any protection from sampling

Protection of tables of magnitude data – 2

1. Identify sensitive cells

- p-percent
- pq
- (n,k)

2. Protect sensitive cells

- Suppression

3. Verify result

- Audit
- Information loss
- (Parameters are not disclosed)

Suppression rules

- Primary suppression rules determine whether a cell could reveal individual respondent information
- Such cells are considered sensitive and cannot be released
- The most common suppression rules are:
 - the p-percent rule
 - the pq rule
 - the (n,k) rule
- These rules are used to identify sensitive cells by verifying whether it is enough difficult for one respondent to estimate the value reported by another respondent too closely

Primary suppression rule: p-percent

- Disclosure of magnitude data occurs if the user can estimate the contribution of a respondent too accurately
- A cell is sensitive, if upper and lower estimates for the respondent's value are closer to the reported value than a pre-specified percentage p
- Formally, a cell is protected if

$$\sum_{i=c+2}^N x_i \geq \frac{p}{100} x_1$$

$x_1, x_2, \dots, x_N$: respondent's value in decreasing order

c : size of a coalition of respondents interested in estimating x_1

- The largest value x_1 is the most exposed

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- The most sensitive value is Alice's, because it is easier to estimate
- If Alice's income cannot be estimated accurately, the income of the other citizens is protected

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice**'s income?

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income?
Bob, Carol, David, whose total income is 130K

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income?
Bob, Carol, David, whose total income is 130K
can estimate that Alice's income is between **80K** and **120K**
 - $\geq 80K$ since it is higher than Bob's
 - $\leq 120K$ ($= 250K - 130K$)

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income?
Bob, Carol, David, whose total income is 130K
can estimate that Alice's income is between **80K** and **120K**
 $\Rightarrow$ sensitive for any $p \geq 20$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income? Bob, Carol, David
- Formally, the cell is protected for any p such that:

$$\sum_{i=c+2}^N x_i \geq \frac{p}{100} x_1$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income? Bob, Carol, David
- Formally, the cell is protected for any p such that:

$$\sum_{i=3+2}^N x_i \geq \frac{p}{100} \text{Alice}$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice**'s income? Bob, Carol, David
- Formally, the cell is protected for any p such that:

$$\sum_{i=5}^N x_i \geq \frac{p}{100} 100$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice**'s income? Bob, Carol, David
- Formally, the cell is protected for any p such that:

$$p \leq \text{Cell} - \sum_{i=1}^4 x_i$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income? Bob, Carol, David
- Formally, the cell is protected for any p such that:
$$p \leq \text{Cell} - (\text{Alice} + \text{Bob} + \text{Carol} + \text{David})$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice**'s income? Bob, Carol, David
- Formally, the cell is protected for any p such that:
$$p \leq 250 - (100 + 80 + 30 + 20)$$

Primary suppression rule: p-percent – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Which is the coalition of $c = 3$ respondents that can better estimate **Alice's** income? Bob, Carol, David
- Formally, the cell is protected for any p such that:
$$p \leq 20$$

Primary suppression rule: pq (1)

- In the p-percent rule, we assumed that there was no prior knowledge about respondent's values
- Agencies should not make this assumption
- In the pq rule, agencies can specify how much prior knowledge there is by assigning a value q which represents how accurately respondents can estimate another respondent's value before any data are published ($p < q < 100$)
- Parameter q represents the error in estimation before the cell is published

Primary suppression rule: pq (2)

- Formally, a cell is **protected** if

$$\frac{q}{100} \sum_{i=c+2}^N x_i \geq \frac{p}{100} x_1$$

$x_1, x_2, \dots, x_N$: respondent's value in decreasing order

c : size of a coalition of respondents interested in estimating x_1

- The pq rule reduces to the p-percent rule when $q=100$ (i.e., no estimate ability)

Primary suppression rule: pq – Example (1)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assume $q=80\%$ to represent how accurately respondents can estimate another respondent's value before any data publishing

Primary suppression rule: pq – Example (1)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assume $q=80\%$ to represent how accurately respondents can estimate another respondent's value before any data publishing
- For **Alice**'s income this implies range **[20,180]**

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- The coalition of $c = 3$ respondents that can better estimate **Alice's** income is Bob, Carol, David

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- The coalition of $c = 3$ respondents that can better estimate **Alice's** income is **Bob, Carol, David**
- The coalition can reduce uncertainty about Alice's income from **[20K-180K]** to **[80K-120K]**
 - $\geq 80K$ since it is higher than Bob's
 - $\leq 120K$ ($=250K - 130K$)

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\frac{q}{100} \sum_{i=c+2}^N x_i \geq \frac{p}{100} x_1$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\frac{80}{100} \sum_{i=3+2}^N x_i \geq \frac{p}{100} \text{Alice}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\frac{80}{100} \sum_{i=5}^N x_i \geq \frac{p}{100} 100$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\frac{80}{100} \sum_{i=5}^N x_i \geq p$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\sum_{i=5}^N x_i \geq \frac{p}{0.80}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$Cell - \sum_{i=1}^4 x_i \geq \frac{p}{0.80}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$\text{Cell} - (\text{Alice} + \text{Bob} + \text{Carol} + \text{David}) \geq \frac{p}{0.80}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$250 - (100 + 80 + 30 + 20) \geq \frac{p}{0.80}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$20 \geq \frac{p}{0.80}$$

Primary suppression rule: pq – Example (2)

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $q = 80\%$ and $c = 3$
- Formally, the cell is protected for any p such that:

$$p \leq 16$$

Primary suppression rule: (n,k)

- Regardless of the number of respondents in a cell, if a small number (n or fewer) of these respondents contribute a large percentage ($k\%$ or more) of the total cell value, the cell is considered sensitive
- Intuitive rule: if a cell is dominated by one respondent, the published total is an upper estimate for her value
- n selected to be larger than the number of any suspected coalitions
- Many agencies use an (n,k) rule with $n = 1$ or $n = 2$

Primary suppression rule: (n,k) – Example

- Consider the respondents that contribute to the total income in a city, which is equal to 250K, to be (in decreasing order)
 - Alice: 100K
 - Bob: 80K
 - Carol: 30K
 - David: 20K
 - Eve: 10K
 - Frank: 3K
 - ...
- Assuming $n=2$ and $k=70$, the cell is considered sensitive
The income of Alice and Bob ($100K+80K=180K$) represents the **72%** of the cell value (250K)

Protection of tables of magnitude data

1. Identify sensitive cells

- p-percent
- pq
- (n,k)

2. Protect sensitive cells

- Suppression

3. Verify result

- Audit
- Information loss
- (Parameters are not disclosed)

Secondary suppression (1)

- Once sensitive cells have been identified, there are two options:
 - **restructure the table and collapse cells** until no sensitive cells remain
 - **cell suppression**: do not publish sensitive cells (primary suppressions) and remove other cells (complementary suppressions)
- An administrative way to avoid cell suppression consists in obtaining written permission from respondents

Secondary suppression (2)

- Other non-sensitive cells must be selected for suppression to ensure that the respondent level data in sensitive cells cannot be estimated too accurately
 - a respondent's data cannot be estimated too closely
- Sensitive cells might be leaked due to the fact that:
 - implicitly published unions of suppressed cells may be sensitive according to the sensitivity rule adopted
 - the row and column equations represented by the published table may be solved, and the value for a suppressed cell estimated too accurately

Secondary suppression (3)

- Any complementary suppression is acceptable as long as the sensitive cells are protected
- For small tables the selection of complementary cells can be done manually
- Data analysts know which cells are of greatest interest (and should not be used for complementary suppression)
- Manual selection of complementary cells is acceptable as long as the resulting table provides sufficient protection to sensitive cells
- An automated audit should be applied to ensure this is true

Protection of tables of magnitude data

1. Identify sensitive cells

- p-percent
- pq
- (n,k)

2. Protect sensitive cells

- Suppression

3. Verify result

- Audit
- Information loss
- (Parameters are not disclosed)

Audit

- If totals are published the sum of the (primary or secondary) suppressed cells can be derived
- Apply the sensitivity rule to these sums to ensure that they are not sensitive
 - Rows and columns can be seen as a large system of linear equations
 - Estimate a lower and upper bound of each suppressed cell using linear programming
 - If bounds are too close to the original value, the cell is sensitive
- Simple for small tables, possibly computationally intractable for large tables

Information loss

- The selection of the complementary cells should result in minimum information loss
- There is no unique definition of information loss
- For instance, we can try to minimize:
 - the sum of the suppressed values (a large number of cells with small values can be suppressed)
 - the total number of suppressed cells

Information in parameter values

While the suppression rules can be published, parameter values should be kept confidential

EXAMPLE: Assume that:

- p-percent rule is used with $p=20\%$ and the same value is used for complementary suppression
- a cell x with value 100 has been suppressed along with other suitable complementary cells
- by solving a system of linear equations, the upper bound is 120 and the lower bound is 80: $80 \leq x \leq 120 \implies x = 100$

Once the value for one suppressed cell has been uniquely determined, other cell values can easily be derived

Protection of tables of magnitude data – Example

Employees by sex and department

Sex	Dept₁	Dept₂	Dept₃	Dept₄	Total
Female	1	2	2	1	6
Male	3	2	0	2	7
Total	4	4	2	3	13

Monthly income by sex and department

Sex	Dept₁	Dept₂	Dept₃	Dept₄	Total
Female	1800	5600	4200	2500	14100
Male	4500	5800	0	5500	15800
Total	6300	11400	4200	8000	29900

Protection of tables of magnitude data – Example

Employees by sex and department

Sex	Dept₁	Dept₂	Dept₃	Dept₄	Total
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(n,k) rule with $n=1$, $k=90 \Rightarrow$ a cell is sensitive if one respondent contributes more than 90%

Protection of tables of magnitude data – Example

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Monthly income by sex and department

Sex	Dept ₁	Dept ₂	Dept ₃	Dept ₄	Total
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(n,k) rule with $n=1$, $k=90 \Rightarrow$ a cell is sensitive if one respondent contributes more than 90%

Protection of tables of magnitude data – Example

Employees by sex and department

Sex	Dept ₁	Dept ₂	Dept ₃	Dept ₄	Total
Female	1	2	2	1	6
Male	3	2	0	2	7
Total	4	4	2	3	13

Monthly income by sex and department

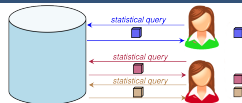
Sex	Dept ₁	Dept ₂	Dept ₃	Dept ₄	Total
Female	D ₁	5600	4200	D ₂	14100
Male	D ₃	5800	0	D ₄	15800
Total	6300	11400	4200	8000	29900

Secondary suppression

Microdata

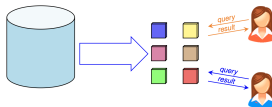
Outline

- Statistical DBMS



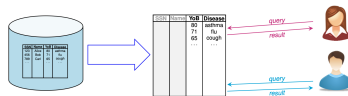
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Macrodata vs microdata

- In the past data were mainly released in tabular form (macrodata) and through statistical DBMS
- Today many situations require that the specific stored data themselves, called microdata, be released
 - increased flexibility and availability of information for recipients
- Microdata are subject to a greater risk of privacy breaches (linking attacks)

Microdata (1)

- Many situations require today that the specific stored data themselves (**microdata**) be released
- The advantage of releasing microdata is an increased **flexibility** and **availability** of information for the recipients
- To protect the anonymity of the respondents, data holders often **remove or encrypt explicit identifiers** such as names, addresses, and phone numbers
- **De-identifying data, however, provides no guarantee of anonymity**

Microdata (2)

- Released information often contains other **quasi-identifying** data (e.g., race, birth date, sex, and ZIP code) that can be linked to publicly available information to reidentify respondents
- The data recipients can determine (or restrict uncertainty) to which respondent some pieces of released data refer
- This has created an increasing demand to devote resources for an adequate protection of sensitive data
- The microdata protection techniques follow two main strategies:
 - **reduce** the information content
 - **change** the data in such a way that the information content is maintained as much as possible

Disclosure risk – Example

SSN	Name	Race	Date of birth	Sex	ZIP	Marital status	Disease
		asian	64/04/12	F	94142	divorced	hypertension
		asian	64/09/13	F	94141	divorced	obesity
		asian	64/04/15	F	94139	married	chest pain
		asian	63/03/13	M	94139	married	obesity
		asian	63/03/18	M	94139	married	short breath
		black	64/09/27	F	94138	single	short breath
		black	64/09/27	F	94139	single	obesity
		white	64/09/27	F	94139	single	chest pain
		white	64/09/27	F	94141	widow	short breath

Name	Address	City	ZIP	DOB	Sex	Status
.....						
.....						
Sue J. Doe	900 Market St.	San Francisco	94142	64/04/12	F	divorced
.....						

Microdata disclosure protection techniques

To limit the disclosure risk, the following procedures should be applied:

- including data from a sample of the whole population only
- removal of identifiers
- limiting geographic details
- limiting the number of variables

Limiting geographic details

- Geographic location is a characteristic that:
 - often appears in microdata
 - can be used for re-identifying respondents
- It is therefore important **limiting geographic details**

EXAMPLE:

- The Census Bureau will not identify any geographic region with less than 100,000 persons in the sampling (250,000 in the '80)
- Microdata contain **contextual variables** that describe the area in which a respondent resides but do not identify that area (e.g., average temperature of an area)

Classification of microdata protection techniques (1)

These techniques are based on the principle that reidentification can be counteracted by reducing the amount of released information:

- masking the data (e.g., by not releasing or by perturbing their values)
- releasing plausible but made up values instead of the real ones

According to this principle, the microdata protection techniques can be classified into two main categories:

- masking techniques
- synthetic data generation techniques

Classification of microdata protection techniques (2)

They can operate on different data types:

- **Continuous.** An attribute is said to be continuous if it is numerical and arithmetic operations are defined on it

EXAMPLE: date of birth, temperature, ...

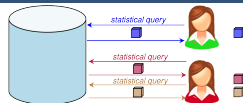
- **Categorical.** An attribute is said to be categorical if it can assume a limited and specified set of values and arithmetic operations do not have sense on it

EXAMPLE: marital status, race, ...

Microdata Disclosure Protection Techniques: Masking Techniques

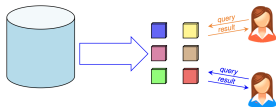
Outline

- Statistical DBMS



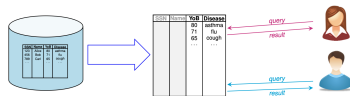
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Masking techniques

- The original data are transformed to produce new data that are valid for statistical analysis and such that they preserve the confidentiality of respondents
- They are classified as:
 - **non-perturbative**, the original data are not modified, but some data are suppressed and/or some details are removed
 - **perturbative**, the original data are modified

Masking techniques – Classification

Non-perturbative

Technique	Continuous	Categorical
Sampling	yes	yes
Local suppression	yes	yes
Global recoding	yes	yes
Top-coding	yes	yes
Bottom-coding	yes	yes
Generalization	yes	yes

Perturbative

Technique	Continuous	Categorical
Resampling	yes	no
Lossy compression	yes	no
Rounding	yes	no
PRAM	no	yes
MASSC	no	yes
Random noise	yes	yes
Swapping	yes	yes
Rank swapping	yes	yes
Micro-aggregation	yes	yes

Sampling

- The protected microdata table is obtained as a **sample** of the original microdata table
- Since there is uncertainty about whether or not a specific respondent is in the sample, reidentification risk decreases

Sampling – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Asian	64/03/09	M	94138	Married	10	190
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/03/18	M	94141	Married	60	290
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/02	M	94138	Single	22	140
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Sampling – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Asian	64/03/09	M	94138	Married	10	190
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/03/18	M	94141	Married	60	290
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
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		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Compute a sample of 3 tuples out of 14

Sampling – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Black	64/09/13	F	94141	Married	15	200
		White	61/09/15	F	94142	Widow	15	200

Local suppression

- It suppresses the value of an attribute (i.e., it replaces it with a missing value) thus limiting the possibilities of analysis
- This technique blanks out some attribute values (sensitive cells) that are likely to contribute significantly to the disclosure risk of the tuple involved

Local suppression – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
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		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Suppress cells that contribute significantly to re-identification

Local suppression – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
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		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
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Suppress cells that contribute significantly to re-identification

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		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F			15	200

Global recoding

- The domain of an attribute is **partitioned into disjoint intervals**, usually of the same width, and each interval is associated with a label
- The protected microdata table is obtained by **replacing** the values of the attribute with the label associated with the corresponding interval

Global recoding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
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		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Global recoding on **Income**:

[150-199]: low, [200-289]: medium, [290-310] high

Global recoding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Global recoding on **Income**:

[150-199]: low, [200-289]: medium, [290-310] high

Global recoding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	med
		Asian	64/09/30	F	94139	Divorced	1	low
		Asian	64/04/18	M	94139	Married	40	med
		Asian	64/04/15	M	94139	Married	17	med
		Black	63/03/13	M	94138	Married	2	low
		Black	63/03/18	M	94138	Married	13	low
		Black	64/09/13	F	94141	Married	15	med
		Black	64/09/07	F	94141	Married	60	high
		White	61/05/14	M	94138	Single	17	low
		White	61/05/08	M	94138	Single	10	high
		White	61/09/15	F	94142	Widow	15	med

Top-coding and bottom-coding

- Top-coding

- It defines an upper limit (**top-code**) for each attribute to be protected. Any value greater than it is replaced with a flag that tells the user what the top-code is and that this value exceeds it
- It can be applied to categorical attributes that can be linearly ordered as well as to continuous attributes

- Bottom-coding

- It defines a lower limit (**bottom-code**) for each attribute to be protected. Any value lower than it is replaced with a flag that tells the user what the bottom-code is and that this value is less than it
- It can be applied to categorical attributes that can be linearly ordered as well as to continuous attributes

Top-coding and bottom-coding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Top-coding on **Holidays** for values higher than 30

Bottom-coding on **Holidays** for values lower than 10

Top-coding and bottom-coding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Top-coding on **Holidays** for values higher than 30

Bottom-coding on **Holidays** for values lower than 10

Top-coding and bottom-coding – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	<10	170
		Asian	64/04/18	M	94139	Married	>30	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	<10	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	>30	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Top-coding on **Holidays** for values higher than 30

Bottom-coding on **Holidays** for values lower than 10

Generalization

- Replaces values with more general values
- Typically based on the definition of a [generalization hierarchy](#), where the most general value is the root and the leaves correspond to the most specific values
- Different generalized microdata tables can be built, depending on the number of generalization steps applied

Generalization – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Generalize attribute **DoB** to the granularity of month

Generalization – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Generalize attribute **DoB** to the granularity of month

Generalization – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09	F	94139	Divorced	13	260
		Asian	64/09	F	94139	Divorced	1	170
		Asian	64/04	M	94139	Married	40	200
		Asian	64/04	M	94139	Married	17	280
		Black	63/03	M	94138	Married	2	190
		Black	63/03	M	94138	Married	13	185
		Black	64/09	F	94141	Married	15	200
		Black	64/09	F	94141	Married	60	290
		White	61/05	M	94138	Single	17	170
		White	61/05	M	94138	Single	10	300
		White	61/09	F	94142	Widow	15	200

Masking techniques – Classification

Non-perturbative

Technique	Continuous	Categorical
Sampling	yes	yes
Local suppression	yes	yes
Global recoding	yes	yes
Top-coding	yes	yes
Bottom-coding	yes	yes
Generalization	yes	yes

Perturbative

Technique	Continuous	Categorical
Resampling	yes	no
Lossy compression	yes	no
Rounding	yes	no
PRAM	no	yes
MASSC	no	yes
Random noise	yes	yes
Swapping	yes	yes
Rank swapping	yes	yes
Micro-aggregation	yes	yes

Random noise

- It **perturbs** a sensitive attribute by adding or by multiplying it with a random variable with a given distribution
- It is necessary to decide whether or not to publish the distribution(s) used to add noise to the data
- Publishing the distribution(s) might increase disclosure risk of the data

Random noise – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Additive noise over attribute **Holidays** (to preserve *average*)

Random noise – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Noise	Income
		Asian	64/09/27	F	94139	Divorced	13	+2	260
		Asian	64/09/30	F	94139	Divorced	1	+1	170
		Asian	64/04/18	M	94139	Married	40	-10	200
		Asian	64/04/15	M	94139	Married	17	+3	280
		Black	63/03/13	M	94138	Married	2	+5	190
		Black	63/03/18	M	94138	Married	13	+8	185
		Black	64/09/13	F	94141	Married	15	+4	200
		Black	64/09/07	F	94141	Married	60	-11	290
		White	61/05/14	M	94138	Single	17	-2	170
		White	61/05/08	M	94138	Single	10	-3	300
		White	61/09/15	F	94142	Widow	15	+3	200

Additive noise over attribute **Holidays** (to preserve *average*)

Random noise – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	15	260
		Asian	64/09/30	F	94139	Divorced	2	170
		Asian	64/04/18	M	94139	Married	30	200
		Asian	64/04/15	M	94139	Married	20	280
		Black	63/03/13	M	94138	Married	7	190
		Black	63/03/18	M	94138	Married	21	185
		Black	64/09/13	F	94141	Married	19	200
		Black	64/09/07	F	94141	Married	49	290
		White	61/05/14	M	94138	Single	15	170
		White	61/05/08	M	94138	Single	7	300
		White	61/09/15	F	94142	Widow	18	200

Additive noise over attribute **Holidays** (to preserve *average*)

Swapping

- A small percent of records are **matched** with other records in the same file, perhaps in different geographic regions, on a set of predetermined variables
- The values of all other variables on the file are then **swapped between the two records**
- This technique reduces the risk of reidentification because it introduces uncertainty about the true value of a respondent's data

Swapping – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Swap **Holidays** and **Income** for tuples with the same **Sex** and **MarStat**

Swapping – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Identify 3 pairs of tuples with same **Sex** and **MarStat**

Swapping – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	2	190
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	40	200
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	60	290
		Black	64/09/07	F	94141	Married	15	200
		White	61/05/14	M	94138	Single	10	300
		White	61/05/08	M	94138	Single	17	170
		White	61/09/15	F	94142	Widow	15	200

Swap **Holidays** and **Income**

Micro-aggregation (blurring)

- It consists in grouping individual tuples into small groups of a fixed dimension k
- The average over each group is published instead of individual values
- Groups are formed by using maximal similarity criteria
- There are different variations of micro-aggregation:
 - the average can substitute the original value only for a tuple in the group or for all of them
 - different attributes can be protected through micro-aggregation using the same or different grouping
 - ...

Micro-aggregation (blurring) – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Group tuples based on **Sex** and **MarStat**

Micro-aggregation (blurring) – Example

SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	260
		Asian	64/09/30	F	94139	Divorced	1	170
		Asian	64/04/18	M	94139	Married	40	200
		Asian	64/04/15	M	94139	Married	17	280
		Black	63/03/13	M	94138	Married	2	190
		Black	63/03/18	M	94138	Married	13	185
		Black	64/09/13	F	94141	Married	15	200
		Black	64/09/07	F	94141	Married	60	290
		White	61/05/14	M	94138	Single	17	170
		White	61/05/08	M	94138	Single	10	300
		White	61/09/15	F	94142	Widow	15	200

Group tuples based on **Sex** and **MarStat**

Micro-aggregation (blurring) – Example

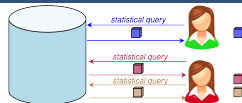
SSN	Name	Race	DoB	Sex	ZIP	MarStat	Holidays	Income
		Asian	64/09/27	F	94139	Divorced	13	215
		Asian	64/09/30	F	94139	Divorced	1	215
		Asian	64/04/18	M	94139	Married	40	213
		Asian	64/04/15	M	94139	Married	17	213
		Black	63/03/13	M	94138	Married	2	213
		Black	63/03/18	M	94138	Married	13	213
		Black	64/09/13	F	94141	Married	15	245
		Black	64/09/07	F	94141	Married	60	245
		White	61/05/14	M	94138	Single	17	235
		White	61/05/08	M	94138	Single	10	235
		White	61/09/15	F	94142	Widow	15	200

Substitute **Income** with the average for each group

Microdata Disclosure Protection Techniques: Synthetic Techniques

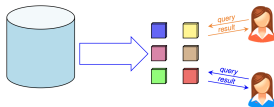
Outline

- Statistical DBMS



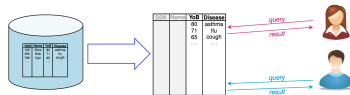
- Macrodata protection

- Count and frequency tables
- Magnitude tables



- Microdata protection

- Masking techniques
- Synthetic techniques



Synthetic techniques (1)

- Since the statistical content of the data is not related to the information provided by each respondent, a model well representing the data could in principle replace the data themselves
- An important requirement for the generation of synthetic data is that the synthetic and original data should present the same quality of statistical analysis
- The main advantage of this class of techniques is that the released synthetic data are not referred to any respondent and therefore their release cannot lead to reidentification

Synthetic techniques (2)

Fully Synthetic

Technique	Continuous	Categorical
Bootstrap	yes	no
Cholesky decomposition	yes	no
Multiple imputation	yes	yes
Maximum entropy	yes	yes
Latin Hypercube Sampling	yes	yes

Partially Synthetic

Technique	Continuous	Categorical
IPSO	yes	no
Hybrid masking	yes	no
Random response	no	yes
Blank and impute	yes	yes
SMIke	yes	yes
Multiply imputed partially synthetic dataset	yes	yes